Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *MENTALCLOUDS* | ***Context:*** *South Asia and MENA NLP/AI Practitioner Cohort — MentalCLOUDS Assessment*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark input is text-only transcriptions of counseling video dialogues [Q14], described as 'incoherent and grammatically unfluent' reflecting natural conversational register [Q61]. Per the dimension priority weights, IF is LOWER priority because both benchmark and the core English-system deployment use case are text-only — no modality mismatch exists for English-interface deployment. The infrastructure context confirms the benchmark's open-source <7B-parameter focus [Q67, Q68] is practically applicable across SA-MENA compute environments. The remaining IF concern is script and right-to-left rendering: the benchmark provides no Arabic, Urdu, or Persian-script input form, but this is more accurately a content/language gap (covered in IC) than a signal-encoding mismatch. For the LOWER-priority IF dimension, this score reflects substantive alignment for English-system deployment with minor gaps for multilingual extension.

ID	Evidence
Q14	“Utilizing pre-processed transcriptions derived from counseling videos, the constituent dialogues within the dataset exhibit a dyadic structure, exclusively featuring patients and therapists as interlocutors.” (p.6)
Q61	“Hence, most of these natural conversations are incoherent and grammatically unfluent.” (p.15)
Q67	“Second, for faster and easier reproduction, we did not assess models larger than 7 billion parameters; however, such models can be part of future examinations.” (p.16)
Q68	“Third, for the initial study and to promote research in this field, only open-source models were assessed in this work.” (p.16)

Strengths

- Text-only modality matches text-only deployment for English-system users; no audio/visual capture infrastructure required [Q14].
- Open-source <7B-parameter model focus [Q67, Q68] aligns with compute constraints in much of the South Asian and MENA research ecosystem.
- Natural conversational register [Q61] matches realistic input distribution for deployed systems.

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Checklist

Checklist item definitions:

ID	Assessment Question

IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: For English-system deployment, signal distributions match — text transcriptions in both source and target. For non-English deployment, the benchmark provides no script-specific input form (RTL Arabic, Nastaliq Urdu, Devanagari, Persian script).

IF-2: Regional infrastructure supports text-based pipelines. Open-source model focus [Q67, Q68] is practically applicable across SA-MENA compute environments. UAE health data localization (Article 13) and Saudi SDAIA constraints affect data movement but not the input-form specification per se.

IF-3: Domain-specific form considerations: counseling transcription quality and turn-taking conventions are encoded in English-language conventions; Arabic diglossia (spoken dialect vs. MSA written) has no analog in the benchmark's input form, but this is principally a content/language issue rather than a signal-encoding mismatch.

IF-4: No documented form mismatches that would harm external validity for the English-system core use case. For multilingual extension, the absence of script-specific input examples is a gap but is downstream of the language-coverage gap addressed in IC and OF.

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WEB-17	UAE health data localization (Article 13 of UAE Health Data Law) constrains cloud-based clinical NLP data pipelines that route data to non-UAE servers (https://www.pwc.com/m1/en/publications/healthcare-data-protection-in-the-uae.html)

Information Gaps

- Specific transcription quality metrics (WER, speaker-segmentation accuracy) are not reported.